

Group 4 — Circuit tracing / attribution

Where this sits. Group 1 (*Landscape*, Part I.3, II.1) showed that an *attention-only* transformer can be rewritten *exactly* into readable linear paths, by freezing the one nonlinearity (the softmax pattern). That rewrite stalls on MLPs: the pointwise nonlinearity does not freeze, and MLPs hold $\sim 2/3$ of the parameters. Group 4 is the engineering and mathematics of getting the rewrite past the MLP — at the cost of trading "exact" for "locally linear" — and then circling back to attack the very thing Group 1 and the rest of Group 4 had *frozen*: the QK bilinear form. Notation follows the *ML Concepts* primer: residual stream as a running sum (§1.3); the attention split $W_{QK} : V \rightarrow V^*$ (a bilinear form, the *where*) and $W_{OV} = W_O W_V : V \rightarrow V$ (an endomorphism, the *what*) (§1.4); transcoders and the cross-layer transcoder (CLT) (§3); virtual weights $W_{\text{in}}^B W_{\text{out}}^A$ (§1.3); and the adjoint W^* (metric-laden) versus dual $W^\vee$ (metric-free) convention.

The shared question

The four papers of this group answer one question: *how do you scale Group 1's exact algebraic rewrite to a real, MLP-bearing frontier model?* The single move that organizes everything is a substitution of error semantics. Group 1's tractability rested on **freezing** attention — which, as the primer stresses (§1.4), is *not* a linearization but an **exact rewriting** of the head's output that happens to be linear in the value-carrying copy of the stream. Group 4 keeps that exact-frozen-attention move but adds two further ingredients to handle the MLP:

1. Replace each MLP by a **cross-layer transcoder** (CLT) — a Group-3-style sparse dictionary model that, crucially, predicts the MLP's *output* from its *input* (it models *computation*, not just representation), so that feature-to-feature interactions "bridge over" the MLP nonlinearity and become linear.
2. Pin the residual difference between the true model and this approximate model, on *one prompt*, with per-(token, layer) **error nodes**.

The result is the **local replacement model**: CLT-for-MLP, frozen attention patterns and frozen normalization denominators, plus error nodes. On the prompt it was built for, it reproduces the underlying model's activations and logits *exactly* (the errors are pinned), yet its only surviving nonlinearities are the feature pre-activations — so it is, as a function of feature activations, an honest linear network. Differentiating it (with stop-gradients on every frozen nonlinearity) yields a **linear attribution graph**: nodes are features, token embeddings, error terms, and output logits; edges are exact linear attributions. This is Ameisen, Lindsey, et al. [2] (the method) and Lindsey et al. [7] (its frontier application).

The two remaining papers attack the part that was frozen. Freezing attention discards the QK circuit — *why* the model attends where it does — which the methods paper itself flags as its most damaging limitation. Kamath et al. [6] and Jermyn et al. [5] **unfreeze the QK bilinear form** and decompose the attention score into feature–feature pairings. The four belong together because they are one pipeline seen at two stages: the OV/MLP attribution that takes attention as given, and the QK attribution that explains the attention.

The papers

Circuit Tracing — the method

Ameisen, Lindsey, et al. [2] fixes the CLT architecture precisely. Write x^ℓ for the underlying model’s residual stream at layer ℓ and y^ℓ for its MLP output. A CLT feature reads at its home layer and writes to *all* later layers:

$$a^\ell = \text{JumpReLU}(W_{\text{enc}}^\ell x^\ell), \quad \hat{y}^\ell = \sum_{\ell'=1}^{\ell} W_{\text{dec}}^{\ell' \rightarrow \ell} a^{\ell'},$$

trained jointly across layers on $L_{\text{MSE}} = \sum_{\ell} \|\hat{y}^\ell - y^\ell\|^2$ plus a decoder-norm-weighted sparsity penalty $L_{\text{sparsity}} = \lambda \sum_{\ell, i} \tanh(c \|W_{\text{dec}, i}^\ell\| a_i^\ell)$. The "cross-layer" reach is a deliberate device: it collapses a feature smeared across several MLP layers (the conjectured *cross-layer superposition*) into one node, which both shortens paths and improves the rewrite. Swapping the CLT in for every MLP gives a **replacement model** that matches the underlying model’s next token on $\sim 50\%$ of (deliberately non-trivial) prompts — a meaningful but partial fit.

Because that 50% gap would compound across layers, single-prompt analysis instead uses the **local replacement model**: CLT-for-MLP, but with attention patterns and normalization denominators *copied* from the true forward pass on the prompt p , and an **error adjustment** at every (token, layer) equal to the residual $y_{\text{true}}^\ell - \hat{y}^\ell$. The authors are explicit about the analogy: this is "similar in spirit to a Taylor approximation of a function f at a point a ; both agree locally... but diverge... as you move away." On p it agrees *exactly* (zero remainder); off p it is only first-order. Viewed as one giant fully connected network spanning all token positions, its weights are exactly the **virtual weights** of the primer’s §1.3 — summed over all linear paths through attention OVs and residual connections, but *not* through MLPs (those are now the CLT) and *not* through the QK pattern (frozen).

The **attribution edge** from source feature s (layer ℓ_s , position c_s) to target t (layer ℓ_t , position c_t) is

$$A_{s \rightarrow t} = a_s \sum_{\ell_s \leq \ell < \ell_t} (W_{\text{dec}, s}^{\ell \rightarrow \ell_t})^* J_{c_s, \ell \rightarrow c_t, \ell_t}^\nabla W_{\text{enc}, t}^{\ell_t},$$

where J^∇ is the Jacobian of the underlying model with a **stop-gradient** on every nonlinearity (MLP outputs, attention patterns, normalization denominators), and a_s is the source activation. (We write the decoder contraction with the adjoint W^* to flag that an inner product on the stream is being used; the architecture supplies none, cf. Part III.) The stop-gradients are what make this exact rather than approximate: with all nonlinearities detached, the pre-activation h_t of any feature is *literally* the sum of its incoming edges, $h_t = \sum_{s \in \mathcal{S}_t} w_{s \rightarrow t}$ — the graph is a genuine linear decomposition of each feature’s activity, not a saliency heuristic. Equivalently, $w_{s \rightarrow t}$ is the derivative of t ’s pre-activation with respect to a_s .

The graph is dense (millions of edges); a **pruning** step keeps the nodes/edges of largest influence, typically cutting nodes $10\times$ while losing $\sim 20\%$ of explained behavior. **Node influence** is computed from the unsigned, input-normalized adjacency matrix A (indexed target $\leftarrow$ source): the indirect-influence matrix sums all path strengths of all lengths,

$$B = A + A^2 + A^3 + \dots = (I - A)^{-1} - I,$$

a **Neumann series**, finite because A is a normalized DAG adjacency (spectral radius < 1). A node’s logit influence is the corresponding row of B averaged over logit nodes. This single operator-inverse is the workhorse: ablation effects correlate with it (Spearman ~ 0.72 for feature–feature

influence) far better than with single-edge attribution. Hypotheses are validated by **constrained patching** — perturbing a feature’s decoding over a chosen layer range while suppressing knock-on effects inside the range, so the intervention isolates the edge the graph claims.

Global weights. Beyond single prompts, the context-independent residual-direct interaction between features s, t is the virtual weight $V_{st} = \langle \sum_{\ell \in L_{st}} W_{\text{dec}}^{s, \ell}, W_{\text{enc}}^t \rangle$ (an inner product of one feature’s accumulated decoders with another’s encoder). Raw V_{st} is dominated by **interference**: with millions of features sharing the stream, pairs that never co-fire still carry large spurious weights (exactly the metric-dependent Gram cross-talk of Part III, and the **interference weights** of Anthropic Interpretability Team [3]). The fix is to fold in coactivation statistics — **target-weighted expected residual attribution**,

$$V_{ij}^{\text{TWERA}} = \frac{\mathbb{E}[a_i a_j]}{\mathbb{E}[a_j]} V_{ij},$$

which reweights by on-distribution co-firing and recovers interpretable connections. The authors are candid that TWERA "is not a perfect solution" and "does not handle inhibition well."

On the Biology of a Large Language Model — the application

Lindsey et al. [7] runs the method on Claude 3.5 Haiku across nine case studies: two-hop reasoning (Dallas $\rightarrow$ Texas $\rightarrow$ Austin, with the intermediate "Texas" step manipulable), planning ahead in rhymed poetry, language-agnostic vs. language-specific multilingual circuits, parallel-heuristic addition (high- and low-precision paths and modular "lookup table" features that constructively interfere), medical-diagnosis chaining, an **inhibitory** "known-answer" circuit whose misfire produces hallucination, a finetuned-in "harmful request" feature, a jailbreak, chain-of-thought (un)faithfulness, and a hidden-goal audit. These are **existence proofs**: each shows the method *can* surface a real, interventionally-confirmed mechanism. The honest headline number is that the graphs give "satisfying insight for about a quarter of the prompts" tried, and that even the success cases capture only a fraction of the model’s computation, with error-node contributions present throughout. The addition case study also exhibits a clean piece of mathematics forced by the architecture: a feature whose pre-activation is affine in the operands obeys a **parallelogram constraint** (if active on $x + y$ and $z + w$ it is active on $x + w$ or $z + y$), so no single feature can compute a general "sum= 5" detector in one step — which is *why* intermediate lookup-table features must exist.

Tracing Attention Feature Interactions — unfreezing QK, part 1

Kamath et al. [6] attacks the frozen half. The premise is exactly the type signature of §1.4: the pre-softmax attention score is a **bilinear function** of the stream at the query and key positions, $\langle W_{QK} x_k, x_q \rangle$ (metric-free, a covector paired with a vector). Decomposing the stream into feature components $x_q = \sum_i a_i^q v_i + \dots$, $x_k = \sum_j a_j^k v_j + \dots$ and multiplying out gives the score as a sum of **feature–feature pairings**

$$\langle W_{QK} x_k, x_q \rangle = \sum_{i,j} a_i^q a_j^k \langle W_{QK} v_j, v_i \rangle + (\text{bias, error}),$$

each term a query-side feature times a key-side feature times their learned QK pairing. This is "QK attribution," and it is exactly the bilinear analogue of the linear OV attribution above. To say *which* heads carry a given attribution-graph edge, the paper adds **head loadings**: forcing

each cross-position edge through attention paths of *length 1* (using residual-stream SAEs / weakly-causal crosscoders to "checkpoint" each layer, sidestepping the exponential explosion of multi-hop attention paths), the attention-mediated edge is $\sum_h a_s a_t (v_t^* O_h V_h v_s) \cdot \text{attn}_h(p_s, p_t)$, a per-head decomposition. Applied to the induction prompt that defeated the methods paper ("... Aunt Sally... Aunt $\rightarrow$ Sally"), QK attribution recovers the mechanism the OV-only graph missed: query-side "Aunt/family-signifier" features pairing with key-side "name" and "this is the name of an aunt" features, validated by steering the key-side feature within the QK circuit.

Progress on Attention — unfreezing QK, part 2

Jermyn et al. [5] reports preliminary work on the harder, structural questions, and is explicitly lab-notes-grade. Three findings matter mathematically. (i) **Attention superposition**: forcing each feature onto a single head (rather than a learned linear combination of heads' patterns) costs 2–4 $\times$ on the L_0 /MSE frontier, evidence that attention features are spread across heads just as residual features are spread across dimensions — superposition recurs one level up. (ii) **Multi-Token Transcoders** (MTCs): a transcoder-like replacement for an *attention* layer, where each feature carries a learned **head-loading vector** (a linear combination of the base heads' patterns, padded with identity to absorb MLP-like local computation); clustering features by head-loading gives semantically coherent families, and the OV dimensionality is a *power-law continuum*, not the expected discrete split between rank-1 skip-trigrams and high-rank copying. The QK and OV conditions appear **coupled** (joint MTC training beats "move SAE features through heads"). (iii) **QK diagonalization**: route query and key through separate encoders, keep only same-index $\langle q_i, k_i \rangle$ interactions, sum and softmax to reconstruct the pattern. Rank-1 QK features struggle to match real patterns at low L_0 , and — tellingly — a small $\mathcal{O}(1000)$ set of induction features does nearly all the work, with the rest dead or sign-cancelling, hinting that faithful QK modelling may need genuinely high-rank features.

Common mathematical core

These papers share a tight kernel of objects, all inherited from earlier groups.

Local linearization (Taylor/Jacobian, exact-at-a-point). The attribution graph is the Jacobian of the local replacement model with stop-gradients. Stop-gradient = holding a nonlinearity's value fixed while zeroing its derivative; doing this on every frozen nonlinearity makes the pre-activation an *exact* linear function of upstream activations *at the prompt*, with the same error semantics as a first-order Taylor expansion: zero remainder at the base point, growing remainder away from it.

Frozen attention, from Group 1. The OV/MLP attribution holds the QK pattern constant, reusing the primer's §1.4 move verbatim — the exact-rewrite-in-content that Group 1 introduced for attention-only models, now generalized (as the methods paper notes the *Framework* itself foresaw) to MLP-bearing models via transcoders.

The QK/OV split as the organizing axis. The whole group is structured by §1.4's type distinction. $W_{OV} : V \rightarrow V$ is an endomorphism: the OV/MLP attributions are *linear*, edges $a_s w_{s \rightarrow t}$. $W_{QK} : V \rightarrow V^*$ is a bilinear form: the QK attributions are *quadratic*, terms $a_i^q a_j^k \langle W_{QK} v_j, v_i \rangle$. Linear vs. bilinear is exactly why the OV side yields a clean graph and the QK side a feature-*pair* table.

Dictionary features, from Group 3. Every node is a transcoder/CLT/SAE feature — a learned overcomplete dictionary direction with a $\mathbb{E}\|\cdot\|^2 + \lambda\rho$ objective (Part III.4). Group 4 is Group 3’s dictionaries wired into a causal graph; the difference is the transcoder’s predict-the-next-activation objective, which buys the linear bridge over the MLP.

Virtual weights and interference, from §1.3 and Group 2. Graph edges *are* virtual weights restricted to non-MLP linear paths; the global-weights interference and its TWERA correction are the §1.3 cross-talk and the Group-2 interference Gram matrix, now causal.

How they diverge

Method vs. frontier application. Ameisen, Lindsey, et al. [2] is a methods paper with quantitative evaluation (replacement/completeness scores, faithfulness metrics) on a controlled 18-layer model plus Haiku; its claims are about the *tool*. Lindsey et al. [7] is a portfolio of *phenomena* on a frontier model, with no completeness guarantees and explicit per-example scoping ("only claims about specific examples"). The mathematics is identical; the epistemic status is not — and the rigour gradient of the corpus (Part V) is visible right here, between the two halves of one method.

Frozen QK vs. unfrozen QK. The deeper fork is internal. Ameisen, Lindsey, et al. [2] and Lindsey et al. [7] *freeze* the QK form and attribute through OV/MLP, gaining exact linearity at the cost of never explaining attention. Kamath et al. [6] and Jermyn et al. [5] *unfreeze* it, gaining an account of attention at the cost of (a) leaving the linear regime for a bilinear one, and (b) trading the CLT’s MLP-bridging linearity for SAE/crosscoder checkpoints that reintroduce nonlinear, prompt-local attributions through the MLP. The two halves cannot both have clean linear attribution *and* explain attention; that tension is the live edge of the program.

What they answer, and what they don’t

What they answer: for a given prompt, *which* features carry the model’s computation along the OV/MLP paths, with intervention-validated influence. What they conspicuously do not:

- **Missing attention circuits.** With QK frozen, the graph explains *that* information moved between positions but not *why*. On induction and multiple-choice prompts this "misses the story" entirely — the methods paper calls such graphs "essentially useless" there. The QK papers begin to fill this in but are preliminary.
- **Reconstruction-error "dark matter".** CLTs reconstruct only part of the MLP; the residual is dumped into **error nodes** that "pop out of nowhere," have no inputs, and are uninterpretable. On off-distribution prompts (e.g. obfuscated jailbreaks) error nodes dominate and the graph reveals little.
- **Mechanistic faithfulness.** Matching input–output behavior is not matching mechanism. One layer past a perturbation the two models agree well (~ 0.8 cosine), but discrepancies **compound across layers**, catastrophically for magnitudes — the exact failure mode Anthropic Interpretability Team [4] (Group 6) constructs in a toy setting and proposes Jacobian-matching to cure.
- **Inhibitory and inactive features.** Attribution is built for "what fired and pushed up"; the fact that a feature *didn’t* fire, and feature-to-feature *inhibition*, are first-class mechanisms (the hallucination circuit is inhibitory) that attribution and TWERA "do not handle well."

- **The ~25% success rate.** On a frontier model the method yields satisfying insight on about a quarter of prompts, and even then on a fraction of the computation. This is an honest measurement, not a footnote.

Connections

To Group 1. This is the scaled-up exact rewrite. Group 1 froze attention and multiplied out an attention-only model exactly; Group 4 keeps the freeze, adds CLTs to clear the MLP, and adds error nodes to restore exactness *on one prompt*. The QK papers literally return to Group 1’s own recommended move — study the QK circuit separately — now with learned features instead of token bigrams.

To Group 3. The nodes are Group 3 dictionaries; the transcoder is the SAE generalized from modelling representation to modelling computation, which is precisely the representation-vs-computation fork of Part IV.2. CLTs and crosscoders are the cross-layer dictionaries (Lindsey et al. [1]).

To Group 6. "Mechanistic faithfulness" is Group 6’s central worry made operational here; the compounding-error measurements are empirical evidence for Anthropic Interpretability Team [4]’s claim that a low-error transcoder can implement the wrong algorithm.

To the three notions of "linear" (Part IV.3). Group 4 is where all three collide in one pipeline. Frozen attention is *exact* (notion 1). The attribution Jacobian is *local linearization* (notion 2). The CLT itself is a *learned sparse approximation* (notion 3), and possibly mechanistically unfaithful. The methods paper’s own Taylor-expansion analogy names notion 2 explicitly; the danger the survey flags is that all three are routinely called "linear" while having different error semantics, and Group 4 stacks them.

For the mathematician

Error semantics: exact rewrite vs. local linearization. Freezing attention (Group 1) replaces $X \mapsto W_{OV}XA(X)^\top$ by $X \mapsto W_{OV}XA_0^\top$ with $A_0 = A(X_0)$: this is an *exact* value at X_0 and globally linear in the content factor — no remainder, ever, as a function of content. Local linearization (the attribution Jacobian) is different: it is the first-order term of a Taylor expansion, exact *only* at X_0 , with remainder $O(\|\Delta X\|^2)$ that the compounding-faithfulness experiments show is not negligible across ~18–40 layers. A mathematician should keep these strictly apart: one is a change of how a fixed map is *written*, the other is a genuine approximation with a derivative-sized domain of validity.

The Neumann-series influence operator. With A the normalized unsigned adjacency of a DAG ($\rho(A) < 1$), $B = (I - A)^{-1} - I = \sum_{k \geq 1} A^k$ is the resolvent minus identity: entry B_{ts} is the total weight of all directed paths $s \rightarrow t$, the path-strength generating function evaluated at 1. This is the standard "total causal effect along all paths" object (cf. Katz / communicability), and it is what makes multi-hop influence tractable: a single linear solve replaces an exponential path sum. Its predictive edge over single-edge attribution (and the role of *path length* via $B_\ell = \sum_{i \leq \ell} A^i$) is empirically the main payoff of the cross-layer design.

The (gauge-)status of an attribution. An attribution edge contracts a decoder vector, a stop-gradient Jacobian, and an encoder covector. The encoder–feature contraction $W_{\text{enc}}x$ is a metric-free pairing (a covector on the stream); the QK score $\langle W_{QK}x_k, x_q \rangle$ is metric-free too. But the *decoder*

side $(W_{\text{dec}})^*(\dots)$ and the cosine/Gram comparisons used to cluster features into supernodes and to compute virtual weights $V_{st} = \langle \sum W_{\text{dec}}, W_{\text{enc}} \rangle$ are metric-dependent (Part III.1): they silently insert $G = I$ on a stream the architecture declares basis-free (§1.3, Part I.4). So an individual edge built from encoder pairings is on firmer footing than the supernode groupings and raw global weights layered on top — TWERA’s reliance on *data* coactivation statistics is, in effect, an empirical substitute for the missing canonical metric.

The combinatorial explosion of QK pairs, and its low-rank rescue. OV/MLP attribution is *linear*: F features give $O(F)$ edges per target. QK attribution is *bilinear*: the score expands into $O(F^2)$ query×key feature pairs per head per position pair — a genuine quadratic blow-up, which the methods paper names as the obstacle to "QK attribution." The structural reason there is any hope is that $W_{QK} = W_Q^\top W_K$ has rank $\leq d_{\text{head}} \ll d_{\text{model}}$: the bilinear form factors through the low-dimensional head space, so the $F \times F$ pairing matrix $\langle W_{QK} v_j, v_i \rangle$ is itself low-rank, and the relevant interactions concentrate in a few directions. Jermyn et al. [5]’s QK diagonalization is exactly an attempt to exploit this — approximate the form by a sum of (ideally rank-1) $\langle q_i, k_i \rangle$ feature interactions — and its difficulty (rank-1 features underfit; a tiny induction-feature set dominates) is direct evidence that the effective rank, while $\ll d_{\text{model}}$, is not 1. The honest open problem is a quantitative low-rank theory of W_{QK} over the feature basis: how the $O(F^2)$ pairs collapse onto an $O(d_{\text{head}})$ -dimensional core, complicated by attention superposition (Jermyn et al. [5]) spreading a single behavior’s QK structure across several heads.

References

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